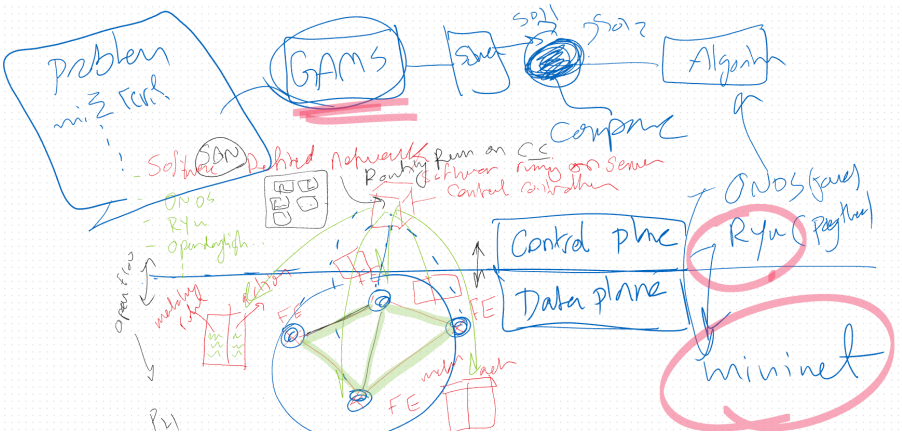
101 - awad's second paper

yaps about his paper: Energy-aware blahblah SDN bender shmender

available in moodle, and his white board available in teams channel



lots of SDN talk.

To achieve these energy savings, he use Software-Defined Networking (SDN): Traditional networks have the decision-making logic baked into every single router. SDN changes this by separating the "Control Plane" (the brain making routing decisions) from the "Data Plane" (the hardware forwarding the data).

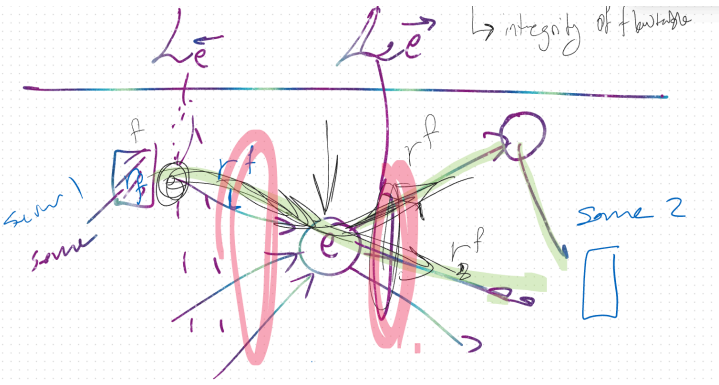
talks about Mininet (a network emulator) and Ryu or ONOS (types of SDN controllers).

- Because the central controller has a view of all traffic, it can make smart, network-wide decisions to pack traffic onto specific paths, allowing unused links to be put to sleep or powered down

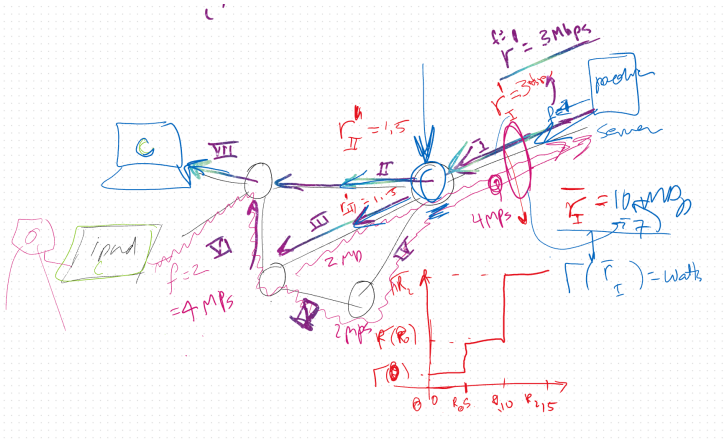
the problem

Similar to first paper, it's a resource allocation problem:
how to significantly reduce the energy consumed by wired computer networks while still reliably delivering all required data.

find Best path s.t.: no congestion, fast delivery, no delay, no packet loss, and integrity of flowtable.



This image shows Flow Conservation: the node "e" and arrows pointing in and out represents "flow conservation." Simply put, the amount of data traffic entering a router must equal the amount of traffic leaving it, unless that router is the final destination.



- This image shows Discrete Link Rates: data flows (like $f = 1$ and $f = 2$) travel across nodes. Look at the red graph in the bottom right. This shows that network links don't consume power linearly; they operate at discrete link rates (like 100 Mbps, 1 Gbps, 10 Gbps), and each jump in speed requires a step up in power (Watts). tl;dr: more speed = more power

The Solution

Calculating the most energy-efficient routes while dealing with these discrete "step" speeds is NP-complete. If you try to solve it perfectly using brute-force math software (dr mentioned "GAMS" and "CPLEX"), it can take hours.

To fix this, he created a Heuristic Algorithm based on a mathematical trick called Benders Decomposition. Instead of trying to solve the massive, complex problem all at once, their algorithm breaks it into smaller, relaxed problems. It quickly finds a near-perfect routing path that minimizes energy consumption while satisfying the data demands.

To achieve "no congestion" and "no packet loss," the algorithm mathematically ensures that the capacity of the activated discrete link rate ($\overline{r}_l$) is always strictly greater than or equal to the actual traffic flowing through it (r_l). By using "integral routing" (sending a whole flow down a single path rather than splitting it up), the algorithm prevents packets from arriving out of order, which helps ensure fast delivery and minimal delay.

What matters in exam

You are given a paragraph like this (but hopefully simpler plz):

II. Network Model and Problem Formulation

We consider an SDN architecture proposed by the Open Networking Foundation (ONF) as shown in Figure 1 [24]. ONF is an organization dedicated to the standardization and promotion of SDN. Three layers constitute this architecture: the infrastructure layer, the control layer and the application layer. The infrastructure layer consists of simple network devices referred to by forwarding elements (FEs). These elements forward packets based on forwarding rules installed in their forwarding tables. In addition, they collect network information and report it to the central controller (CC). The control layer serves as a liaison between the application and infrastructure layers through its northbound and southbound application programming interfaces (APIs). One of the first southbound APIs is OpenFlow [24]. The application layer consists of network applications designed to control and optimize the network. Examples of these applications are routing algorithms, resource allocation schemes, load

ACCEPTED MANUSCRIPT

balancing schemes, mobility management schemes and intrusion detection systems access. Network applications access a global view of the network which is maintained by the CC to control and optimize the network operation. The network application control instructions are represented by flow rules and pushed by the CC to FEs' forwarding tables.

need to turn it into a set of defined variables:

The network is formed by a single CC and a finite number of FEs. The CC computes the optimal integral route for each demand based on the proposed solution and pushes the results to the FEs, which implement these routes as flow rules in their forwarding tables [25]. The network is modeled by an undirected graph $G(\mathcal{E},\mathcal{L})$, where $\mathcal{E}=\{1,\cdots,e,\cdots,E\}$ is the set of FEs and $\mathcal{L}=\{1,\cdots,l,\cdots,L\}$ is the set of bi-directional links connecting them. $E=|\mathcal{E}|$ and $L=|\mathcal{L}|$ denote the cardinality of the set of FEs and set of links. The set $\mathcal{L}$ does not include links connecting the FEs to the central controller. The subset of links $\mathcal{L}$ originated at FE e is denoted by $\mathcal{L}_e^{\rightarrow}$, whereas the subset of links terminated at e is denoted by $\mathcal{L}_e^{\leftarrow}$; their union is $\mathcal{L}_e$. There is a set of F data flows, represented by $\mathcal{F}=\{1,\cdots,f,\cdots,F\}$, to be routed through the network. The origin and destination FEs of the fth flow are represented by $o(f)$ and $d(f)$, respectively. A given flow $f\in\mathcal{F}$ is routed from $o(f)$ to $d(f)$ on a path $\mathcal{P}^f$, which is the set of connected links forming the route of f. It requires a rate $r^f>0$ on each link $l\in\mathcal{P}^f$.

The data rate required to support a given flow f is denoted by r^f. Let $r_l^f\geq 0$ be the decision variable reflecting the data rate required on link $l\in\mathcal{L}$ by flow $f\in\mathcal{F}$. When $l\in\mathcal{P}^f,\ r_l^f=r^f$. By contrast, when f does not travel through $l,\ r_l^f=0$. The flow rates through an FE $e,\ e\in\mathcal{E}$, must satisfy the flow conservation constraints. For every flow $f,\ f\in\mathcal{F}$, and FE $e,\ e\in\mathcal{E}$,

$$\sum_{l\in\mathcal{L}_e^{\rightarrow}} r_l^f-\sum_{l\in\mathcal{L}_e^{\leftarrow}} r_l^f=\begin{cases}0 & e\neq o(f) \text{ and } e\neq d(f)\\ r^f & e=o(f)\\ -r^f & e=d(f).\end{cases} \tag{1}$$

Consequently, the sum of the rates routed through a given link $l,\ l\in\mathcal{L}$, can be written as

$$r_l=\sum_{f\in\mathcal{F}} r_l^f. \tag{2}$$

These rates constitute a vector $\mathbf{r}=[r_l]_{l\in\mathcal{L}}$ of size L.

To support the routed flow r_l through link $l,\ l\in\mathcal{L}$, the CC activates one of the available discrete link rates $\overline{r}_l\in\overline{\mathcal{R}}=\{R_0,R_1,\cdots,R_{\max}\}$, where the available rates are sorted in ascending order (i.e., $R_0<R_1<\cdots<R_{\max}$). Stated differently, the activated link rate $\overline{r}_l$ for link $l,\ l\in\mathcal{L}$, is given by

$$\overline{r}_l=\begin{cases}R_0 & 0<r_l\leq R_0,\\ R_1 & R_0<r_l\leq R_1,\\ \vdots & \vdots\\ R_{\max} & R_{\max-1}<r_l\leq R_{\max}\end{cases}. \tag{3}$$

An activated discrete operating rate $\overline{r}_l\in\overline{\mathcal{R}}$ for a link $l,\ l\in\mathcal{L}$ engenders energy consumption

$$\Gamma_l(\overline{r}_l)=\begin{cases}\gamma_0 & \text{if } \overline{r}_l=R_0,\\ \gamma_1 & \text{if } \overline{r}_l=R_1,\\ \vdots & \vdots\\ \gamma_{\max} & \text{if } \overline{r}_l=R_{\max}\end{cases}. \tag{4}$$

The set of discrete rates activated on the L links in the network are denoted by the vector $\overline{\mathbf{r}}=[\overline{r}_l]_{l\in\mathcal{L}}$ of size L.

Based on the above description, the problem of routing flows and setting discrete link rates consists of finding the flow rates $r_l^f,\ l\in\mathcal{L}, f\in\mathcal{F}$ that minimize network energy consumption. Formally, the problem is equivalent to the following

Accepted Manuscript

so you can finally turn it into a math form with constraints:

$$\min \sum_{l=1}^{L} \Gamma_l(\overline{r}_l) \tag{5a}$$

$$\text{subject to} \quad r_l = \sum_{f \in \mathcal{F}} r_l^f \qquad l \in \mathcal{L} \tag{5b}$$

$$\sum_{l \in \mathcal{L}_e^{\rightarrow}} r_l^f - \sum_{l \in \mathcal{L}_e^{\leftarrow}} r_l^f = \begin{cases} 0 & f \in \mathcal{F}, e \in \mathcal{E}, e \neq o(f), e \neq d(f) \\ r^f & f \in \mathcal{F}, e = o(f) \\ -r^f & f \in \mathcal{F}, e = d(f). \end{cases} \tag{5c}$$

$$\overline{r}_l = \begin{cases} R_0 & 0 < r_l \leq R_0, \\ R_1 & R_0 < r_l \leq R_1, \\ \vdots & \vdots \\ R_{\max} & R_{\max-1} < r_l \leq R_{\max}. \end{cases} \quad l \in \mathcal{L} \tag{5d}$$

The objective function in (5a) minimizes energy consumption at all links. The constraint (5b) defines the link rate required to support all flows passing through it. Flow conservation is guaranteed in constraint (5c). The discrete link rate corresponding to the sum of flows through link l is defined in constraint (5d).The mathematical formulation in (5) is equivalent to a mixed-integer programming problem with a non-convex discrete-cost step increasing function. The integral routing constraint and discreteness of the objective function make the problem NP hard [26].

Manuscript

gemini explanation of the math slop soup:

For your task of translating a resource-allocation problem into its mathematical form, the most relevant part of this paper is Section II: Network Model and Problem Formulation. Specifically, the original optimization problem is encapsulated in Equations (5a) through (5d).

Here is the exact mathematical formulation extracted from the paper:

Objective Function

The goal of the problem is to minimize the total energy consumption across all links in the network:

$$min \sum_{l=1}^{L} \Gamma_l(\overline{r}_l)$$

Constraints

The problem is bound by three main constraints:

- Total Link Rate: The rate on any given link l must equal the sum of the rates of all flows f utilizing that link:

$$r_l = \sum_{f \in \mathcal{F}} r_l^f \quad \forall l \in \mathcal{L}$$

- Flow Conservation: This ensures that traffic entering a forwarding element equals the traffic leaving it, unless the element is the origin or destination of the flow:

$$\sum_{l \in \mathcal{L}_e} r_l^f - \sum_{l \in \mathcal{L}_e^-} r_l^f = \begin{cases} 0 & f \in \mathcal{F}, e \in \mathcal{E}, e \neq o(f), e \neq d(f) \\ r^f & f \in \mathcal{F}, e = o(f) \\ -r^f & f \in \mathcal{F}, e = d(f) \end{cases}$$

- Discrete Link Rate Allocation: The continuous required rate r_l must be mapped to one of the available discrete operating rates $\overline{r_l}$ provided by the hardware:

$$\overline{r_l} = \begin{cases} R_0 & 0 < r_l \leq R_0 \\ R_1 & R_0 < r_l \leq R_1 \\ \vdots & \\ R_{max} & R_{max-1} < r_l \leq R_{max} \end{cases} \qquad \forall l \in \mathcal{L}$$

(Note: Later in Section III, the authors rewrite this into a relaxed "Benders decomposition" format to actually solve it, but if you only need the core mathematical representation of the problem itself with its direct constraints, Equations 5a-5d are exactly what you are looking for).